

State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

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11

retrieval branches

146

ranker features

0.3811

Blind-B composite

29_{/40}

where that finished

The task

SETUP **OPTIONAL**

EVERY ASSISTANT TURN

- Retrieve **20 tracks** from a shared **47,071-track** catalog
- Write a short reply presenting the top recommendation

PER-TRACK METADATA

Title, artist, album, a median of **17** genre/mood tags, popularity, release date, plus precomputed text, audio, cover-art and collaborative-filtering embeddings.

SPLITS

Train	15,199 sessions (~121.6k turns)
Development	1,000 sessions × 8 turns = 8,000
Blind-A / Blind-B	80 sessions each, final turns

Blind-B additionally withholds the conversation-goal annotations present in training data.

BUILT ON

TalkPlayData 2. The conversations are **LLM-generated**, not observed listening sessions — which matters later.

The composite metric

SETUP OPTIONAL

$$0.50 \cdot \text{nDCG@20} + 0.10 \cdot \text{cat.div} + 0.10 \cdot \text{lex.div} + 0.30 \cdot (\text{judge} - 1)/4$$

OUR BLIND-B DECOMPOSITION

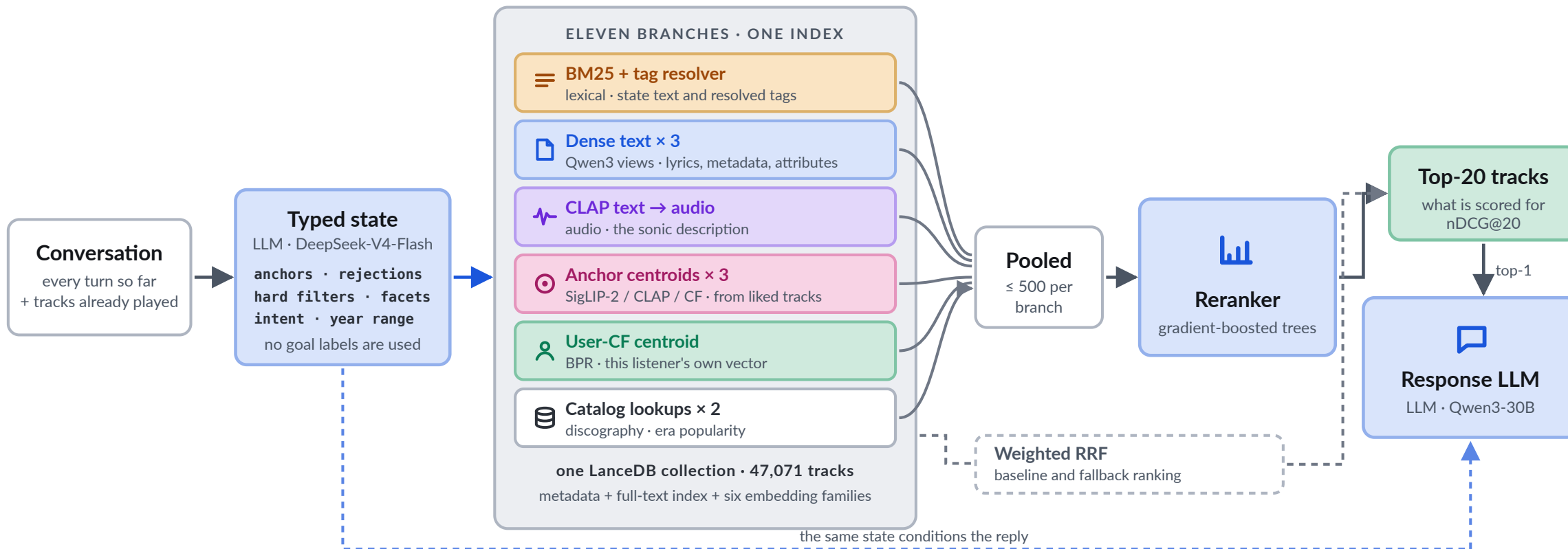
$0.50 \times \text{nDCG@20}$	0.2537	0.1269
$0.10 \times \text{cat. div.}$	0.0315	0.0032
$0.10 \times \text{lex. div.}$	0.7862	0.0786
$0.30 \times (3.30-1)/4$		0.1725
Composite		0.3811

WHERE THE SCORE CAME FROM

The **judge term** out-earned **nDCG**, and category diversity returned **0.003** of the 0.10 on offer. We optimised almost entirely for relevance and left the diversity terms alone.

One state, eleven branches, one learned ranker

APPROACH **CORE**



State extraction

APPROACH **OPTIONAL**

THE EXTRACTOR

DeepSeek-V4-Flash (build 0423), schema-constrained, given the dialog so far and the IDs of tracks already played.

DELIBERATELY BLIND

No conversation goals, goal progress, or user profiles reach the extractor. The state is whatever an LLM can read off the conversation.

WHY THIS MODEL

Cheaper extractors under-filled the optional fields — rejections and facets especially — so every final run uses this one. Per-turn extraction is a paid API call, which is what kept us off the training split.

PROMPT VERSION

ConversationStateV1, projected to VOPlus for the downstream contract.

What the state can hold

APPROACH **OPTIONAL**

THE REQUEST

- request type
- intent mode
- routing flags

WHAT THEY LIKE

- anchors and pinned tracks
- per-track feedback
- mentioned entities

WHAT THEY RULED OUT

- rejected tracks
- rejected artists
- hard filters

WHAT IT SOUNDS LIKE

- mood facets
- genre facets
- sonic facets

WHAT IT IS ABOUT

- lyrical theme

BOUNDS

- release-year window
- process constraints

State extraction on a real Blind-B turn

APPROACH **OPTIONAL**

User (turns 2–3): "...subdued, female singer, a sense of place in the lyrics, stark almost a cappella delivery... not it either."

Extracted state (abridged):

request_type: hidden_target
anchor artist: Neko Case (must_use)
rejected tracks: "Calling Cards", "Man"
facets: mood=subdued; lyrical_theme=sense of place;
 sonic=stark almost a cappella

Top-1: Neko Case – "Bracing For Sunday"

WHERE EACH FIELD GOES

- The **anchor** drives the centroid searches
- The **facets** become text and audio queries
- The **rejections** filter the fused pool

AND WHERE THEY DO NOT

Nothing here becomes a hard filter on album, and the year window never reaches the index. That gap is the main diagnostic later.

Six request types, six shapes of state

APPROACH OPTIONAL

REQUEST TYPE	A REAL TURN	WHAT THE EXTRACTOR PULLED OUT
<code>exact_track</code>	"I wanted <i>Watercolors</i> by Pat Metheny specifically."	track = <i>Watercolors</i> must_use · artist = Pat Metheny must_use
<code>exact_album</code>	"Which song from Peter Doherty's <i>Grace/Wastelands...</i> "	artist + album must_use · facet "best tracks of 2009"
<code>same_artist</code>	"Got another track known for his guitar work?"	artist must_use · prior track do_not_use · facets: guitar work, melodic complexity
<code>new_artist</code>	"...branching out to other singer-songwriter artists."	artist = Ryan Adams & The Cardinals do_not_use · facets: acoustic, introspective
<code>attribute_search</code>	"...a really striking, abstract cover art, bold colors and shapes."	three visual facets · routing flag image_or_visual_search
<code>hidden_target</code>	"A Lil Wayne song from an experimental phase."	artist must_use · facet "experimental" · flag hidden_target_search

The extractor read all six of these correctly. Across the whole Blind-B run only **3 of 80 turns** were diagnosed as a state-extraction miss — reading the conversation was almost never the problem.

Routing flags decide which branches fire

APPROACH **OPTIONAL**

FLAG	OPENS
exact_entity_probe	catalog lookups — discography, album
lyric_search	the Qwen3 lyrics view
feature_articulation	CLAP text→audio on the sonic description
image_or_visual_search	the gated SigLIP-2 cover-art branch
hidden_target_search	widens the pool; suppresses exact pins

A REAL VISUAL REQUEST

“I'm looking for an album with a really striking, abstract cover art... bold colors and shapes, almost like a modern painting.”

→ three visual facets, `image_or_visual_search = true`

AND THE HONEST PART

The reranker carries **no features** for that branch. Even when the flag opens it, the ranker cannot weigh what it returns.

Grounding names to the catalog

APPROACH **OPTIONAL**

NAMES → IDS

Fuzzy string matching of artist and track names against the catalog name lists, keeping matches above **80/100**. Downstream stages only ever see identifiers.

FREE TEXT → TAGS

- 1 Exact match against the tag vocabulary
- 2 Alias match
- 3 Substring match
- 4 Embedding nearest neighbour, cosine ≥ 0.6 , top 3

Used by the BM25 branch. Outside it, tag matching falls back to exact strings — part of why so little of the state became executable.

The eleven retrieval branches

APPROACH OPTIONAL

BRANCH	SPACE	QUERY
BM25 (fielded)	lexical	state text + resolved tags
Dense text × 3	Qwen3 views	state-built strings
CLAP text→audio	audio	sonic description
Anchor centroids × 3	SigLIP-2/CLAP/CF	liked-track centroids †
User-CF centroid	CF-BPR	the listener's own vector
Lookups × 2	catalog	discography; era popularity

GATED TWELFTH

SigLIP-2 text→cover-art fires only on a visual request. The reranker carries no features for it.

POOL DEPTH

Each branch truncated to its top 500 before the union reaches the ranker.

† Gated by intent mode: a pivot suppresses anchor centroids.

One LanceDB collection

APPROACH **OPTIONAL**

COLUMN	MODEL	DIM
Lyrics text	Qwen3-Embedding-0.6B	1024
Metadata text	Qwen3-Embedding-8B	—
Attributes text	Qwen3-Embedding-8B	—
Audio	LAION-CLAP	512
Cover art	SigLIP-2	768
Collaborative filtering	BPR	128
Bi-encoder document	b1 (fine-tuned Qwen3-4B)	2560

THE INTENT

Express every filter in one space, so retrieval could in principle collapse into a single composed query.

WHAT HAPPENED

We never got there. Branches stayed separate and most state fields never became query predicates at all.

Weighted RRF — baseline and fallback

APPROACH **OPTIONAL**

FUSION

Weighted reciprocal-rank fusion, $k = 60$, top-1000 across the branch pools.

POST-FUSION ADJUSTMENTS

- Drop rejected artists and tracks
- Drop tracks already played
- Promote / demote on exact tag membership
- Year-range decay

AS A BASELINE

0.1492

devset nDCG@20, no training

STILL THE FALLBACK

If the learned ranker is unavailable, this order ships. It is the only part of the ranking stack with no trained component.

What the ranker leans on

APPROACH **CORE**

FEATURE FAMILY	#	GAIN
Bi-encoder cosine	1	59.0%
Per-branch rank and score	66	11.5%
Session and artist affinity	18	10.7%
Popularity · state · tags	31	9.2%
Other similarity cosines	17	8.9%
Constraints and rejections	13	0.7%

SETUP

LightGBM LambdaMART, a gradient-boosted tree ranker.
LR 0.025, 127 leaves, truncation 200, 5-fold user-grouped CV. ~20.6M candidate rows.

READ IT CAREFULLY

Split gain describes the fitted model, not recommendation quality. But **constraints at 0.7%** is real: the ranker leans on similarity, not on what the listener asked for.

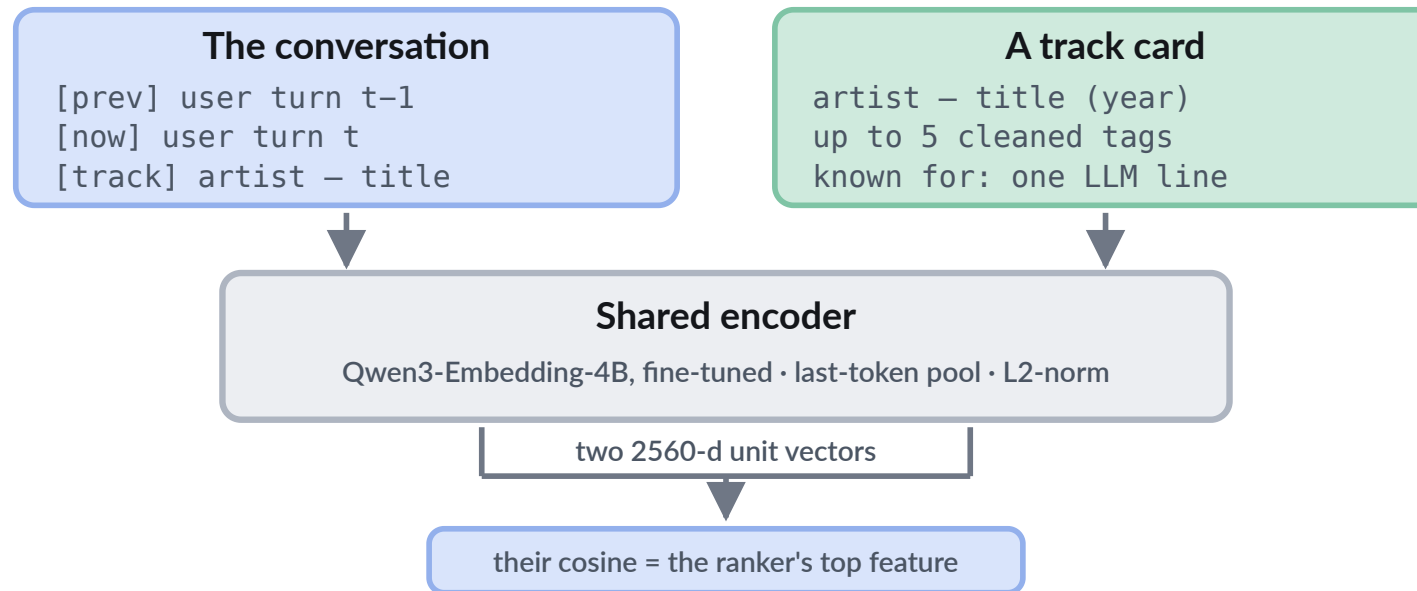
GUARDS AFTER SCORING

Exact-track pin when a specific track is named; a final artist-constraint check.

The bi-encoder behind that top feature

APPROACH

OPTIONAL



How b1 was trained

APPROACH **OPTIONAL**

$$L = -(1/B) \sum_i \log \left[\exp(v_{q_i}^T v_{d_{i+}} / \tau) / \sum_{d \text{ in } D_i} \exp(v_{q_i}^T v_d / \tau) \right] \quad \tau = 1/20$$

POSITIVES

53,885

(turn, track) pairs whose *next* turn's goal-progress says the recommendation moved the session forward

NEGATIVES

4 + batch

four mined hard negatives per positive, plus every in-batch document

RETRIEVAL VS RANKING

As a twelfth *retrieval* branch it added little to the union's coverage and was dropped. As a *ranking* feature it dominates the fit — out-of-fold nDCG@20 0.1970 → 0.2032.

Not isolated: which parts of the rendering matter, and whether the effect carries to the blind splits.

Down-weighting contradicted turns

APPROACH **OPTIONAL**

WEIGHT, BY WHAT THE NEXT TURN SAYS

User rejects the labelled track	× 0.3
Goal-progress says it did not help	× 0.3
User rejects only its artist	× 0.6
Floor	0.2

Chosen by hand. Never tuned.

THE GOAL-ANNOTATION ASYMMETRY

Goal-progress annotations shape **training labels** — these weights, and the bi-encoder's positives — but are never **features**. Blind-B removes them, so the deployed ranker is goal-free at inference.

Response generation – 30% of the composite

APPROACH **OPTIONAL**

THE DEPLOYED STYLE INSTRUCTION

“Write 1–2 concise sentences about only the selected track. Prioritize the latest user request and extracted state over older conversation history. If the track is reasonably aligned, explain the fit with one specific supported reason. If it clearly conflicts with an explicit avoid/new-artist constraint, do not oversell it or blame the system; briefly frame the limitation and the closest supported reason.”

SETUP

Qwen3-30B-A3B-Instruct-2507, temperature 0. One pass over the top-1 track as a delimited XML block (≤ 10 tags) plus the **latest** state.

HOW THE PROMPT WAS CHOSEN

Eight variants swept over a frozen Blind-A retrieval output → judge **4.70**, no recommendations changed. Selected on the leaderboard score.

MISSING

No multi-draft sampling, no selection, no checking, no repair pass.

How it scored

RESULTS **CORE**

OFFICIAL CODABENCH SPLIT	NDCG@20	CAT.	LEX.	JUDGE	COMPOSITE
Blind-A • development phase	0.4380	0.0313	0.8028	4.70	0.5799
Blind-B • final, 29th of 40	0.2537	0.0315	0.7862	3.30	0.3811

NDCG@20

-42%

0.4380 → 0.2537

JUDGE (1-5)

-1.40

4.70 → 3.30

DIVERSITY TERMS

flat

category and lexical barely moved

Blind-B also withholds the conversation-goal annotations — but the deployed ranker never used them at inference, so that is not the explanation.

Development split — what we steered on

SYSTEM (HOW MEASURED)	NDCG@20
Weighted RRF fusion (no training)	0.1492
LambdaMART, out-of-fold CV	0.1970
+ b1_cos (deployed feature set)	0.2032
Historical v10 LambdaMART, in-sample	0.3844
Official Blind-B	0.2537

DO NOT OVER-READ THE FIRST GAP

The fusion and learned-ranker configurations used **different candidate-pool depths**, so 0.149 vs 0.197–0.203 does not isolate the effect of learned re-ranking.

THE IN-SAMPLE ROW

A historical goal-aware model without b1_cos. Not directly comparable — shown because it is the number we were selecting on.

What is not ideal

We tuned on what we measured.

The 8,000-turn development set was both our training and our evaluation data, and the response prompt was picked against the Blind-A leaderboard: 4.70 there, 3.30 on Blind-B.

We extracted more than we used.

The state we read off a conversation was far richer than the state that reached retrieval or ranking. Most fields never became a filter or a ranking signal at all.

Nothing from outside the catalog.

No outside source for what a track or artist is known for, and no tempo, key or soundtrack. Past one CF vector, no listening behaviour — no co-occurrence, no transitions.

The reply was never checked.

Asked for a track we do not have, the generator described a different song by the same artist instead of saying so.

Asked for, served

ANALYSIS **CORE**

the Kamelot track
from **Silverthorn**



Kamelot – Fallen Star, from **Haven**
CONSTRAINT NEVER APPLIED

exactly **126.70 bpm**,
in G minor



no tempo or key field exists to match on
NOTHING IN THE CATALOG

“Watercolors”
by Pat Metheny



Metheny – Alfie, described as if it were
RESPONSE LLM

Silverthorn is in the catalog. The third is a response failure, not a retrieval one – the track simply does not exist in the 47,071.

Where the failures were — all 80 Blind-B turns

ANALYSIS OPTIONAL

DIAGNOSED GAP	TURNS	MEANING
Ranking	31	a better, non-violating candidate sat below our top-20
Constraint not applied	8	an extracted filter never reached the pool
State extraction	3	the extractor missed something the user said
Entity resolution	1	a name never resolved to a catalog ID

TOP-1 FLAGGED

17 / 80

the track we led with violated something the state had recorded

CATEGORIES OVERLAP

43 diagnoses across 32 distinct turns. This is a heuristic audit, not a relevance judgment — Blind-B exposes no labels.

Extraction is 3 of 43 diagnoses. The failure is overwhelmingly downstream of understanding the conversation.

Failure 1 — the constraint was extracted, then ignored

ANALYSIS **CORE**

USER *"...I gotta break away from Deltron and Del completely. Looking for **new** established underground hip-hop artists."*

STATE avoid **Deltron 3030**, **Del The Funky Homosapien** · request_type **new_artist**

WE SERVED **Dead Prez – Hip Hop** (classic hip-hop, the thing they rejected)

USER *"...branching out to other singer-songwriter or alt-country artists."*

STATE avoid **Ryan Adams & The Cardinals** · request_type **new_artist**

WE SERVED **Ryan Adams – When The Stars Go Blue**

USER *"...still focusing on Juanes. While I enjoy his music, I haven't found the specific lyrical depth I'm after."*

STATE rejected **Juanes** · avoid **Juanes**, **Latin Pop**

WE SERVED **Juanes – Me Enamorado De Ti**

Three different sessions, one failure: the listener asked to move on, the state recorded it, and **the ranker served that artist anyway**. Constraint features carry 0.7% of split gain — nothing in the model was weighing them.

Failure 2 — nothing in the data to match

ANALYSIS **OPTIONAL**

USER *"I wanted Watercolors by Pat Metheny specifically. Can you play that one?"*
STATE track **Watercolors** must_use · artist **Pat Metheny** must_use – read perfectly
REALITY the track is not in the 47,071-track catalog
WE SERVED **Pat Metheny – Alfie, described as if it were the one asked for**

USER *"Exactly 126.70 bpm, in G minor."*
REALITY no tempo field, no key field – no branch can match on either

USER *"A song from the Breaking Bad soundtrack."*
REALITY no film or TV metadata anywhere in the catalog

Extraction was correct every time. The gap is **knowledge, not understanding** — and when the answer does not exist, the system should say so rather than offer the nearest thing in its place.

Failure 3 — the better track was already in our pool

ANALYSIS OPTIONAL

WHAT THEY ASKED FOR	WE SERVED AT RANK 1	ALREADY IN OUR POOL	POOL RANK
nu-jazz, downtempo, female vocals — <i>not</i> Parov Stelar	Parov Stelar — Booty Swing	St Germain — What You Think About...	34
new underground hip-hop, not Deltron	Dead Prez — Hip Hop	Souls Of Mischief — Make Your Mind Up	31
hardcore, not classic hip-hop	Dead Prez — Hip Hop	Papa Roach — Getting Away With Murder	49
that carol, but not Beegie Adair again	Beegie Adair — The Christmas Song	Frank Sinatra — Have Yourself A Merry Little Christmas	63

On **31 of 80 Blind-B turns** a non-violating, better-fitting candidate was sitting in our own pool below the top-20 we served. Retrieval found it; ranking buried it.

Auditing the submitted run

SINGLE-JUDGE QUALITY RATING

78%

of Blind-B turns rated weak-or-bad by one DeepSeek-V4-Flash judge

READ AS

Diagnostic, not ground truth. Blind-B exposes no relevance labels, so only the validation failure is actually *measured*.

DETERMINISTIC TRACE AUDIT – 43 DIAGNOSES OVER 32 TURNS

Heuristic alternative below top-20	31
Under-applied constraint	8
State-extraction miss	3
Entity-resolution miss	1

Categories overlap. The alternative was chosen by positive-term metadata matches plus popularity – an audit heuristic, not a relevance judgment.

The target often ignores the request

ANALYSIS **CORE**

User: "...something with a similar chill, electronic vibe, but **from a different artist?**" (just played: Bonobo)

Target: **Bonobo – "Jets"** · annotated MOVES toward goal

Next turn: "'Jets' is cool, but I was actually hoping for something **from a different artist this time...**"

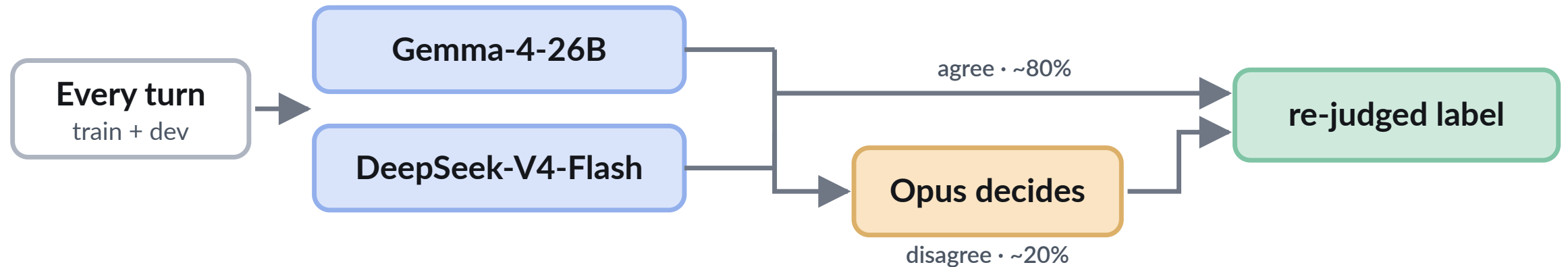
Target: **Bonobo – "Pieces"** · annotated MOVES toward goal

Ground truth is simply **whatever played next**, in a session an LLM wrote. Our own model anchors on the just-played artist — we think it learns that here.

How the LLM relabelling worked

ANALYSIS **OPTIONAL**

Both judges score the same two axes: was a different artist asked for, and does the track fit the request?



What the relabelling found

ANALYSIS **OPTIONAL**

RE-JUDGED

113,393

labelable train + dev turns

NOT SETTLED AS A GOOD FIT

61%

of the 106,393 training turns

OUTRIGHT ANCHORING

18,222

17% — the label keeps the rejected artist

WHAT HAPPENED WHEN WE USED THEM

A ranker trained on the re-judged labels scored **lower** — so the submission kept the originals.

THE CATCH

That score is computed **against the original labels**, so it could not have favoured the relabeling. We release the re-judged labels as evidence to audit, not as a corrected benchmark.

Limits of our own analysis

ANALYSIS OPTIONAL

- The relabeling was **never checked against human judgments**.
- The Blind-B quality audit used a **single** LLM judge.
- The trace audit was **heuristic**, not a relevance judgment.
- The conversations are **LLM-generated**, not observed listening sessions.
- Neither blind split exposes relevance labels — blind-set failure rates and relabeling rates are **diagnostic, not ground truth**.
- Development-split and leaderboard numbers are **separate measurements**.

What we would do differently

CONCLUSION **CORE**

- 1 **Hold out a clean split before touching anything.** Free, and it would have changed every selection decision we made.
- 2 **Make the state executable.** Rejections and constraints carried 0.7% of model gain — most of what the extractor understood never became a filter.
- 3 **Import knowledge the catalog lacks** — tempo, key, soundtrack and film/TV links — so requests naming them have something to match.
- 4 **Check the response against the state** before emitting it: flag unavailable tracks and unmet constraints instead of overselling a substitute.
- 5 **Decide what to do when the benchmark and your judgment disagree.** We still do not have a principled answer.

Everything is released

CONCLUSION **CORE**

IN THE REPO

- Pipeline code, configs, trained models
- Cached per-turn extraction states
- The 113,393 re-judged labels
- A frozen-LLM-cache replay bundle

ZERO-CREDENTIAL REPLAY

Every LLM call is served from a curated frozen cache that **hard-fails on a miss** rather than falling through to a paid call. Both submissions reproduce without any API key.



Code, models, data

[github.com/npatta01/
music-crs-2026](https://github.com/npatta01/music-crs-2026)

Appendix

TYPE A NUMBER + ENTER · OR G TO SEARCH

1 Related work

2 Reproducing the run

Related work — four established lines

A1

CONVERSATIONAL STATE TRACKING

CRS systems commonly track dialog state explicitly and condition retrieval on it. We follow that, but compile the state with a **prompted LLM** instead of a trained tracker.

MODALITY-SPECIFIC MUSIC ENCODERS

Contrastive language-audio pretraining (CLAP), vision-language over cover art (SigLIP-2), collaborative filtering (BPR), general text embeddings (Qwen3).

RETRIEVE-THEN-RERANK AT CATALOG SCALE

Pool multiple sources with reciprocal-rank fusion; rank with LambdaMART.

OUR CLAIM

What we report is **how that combination behaved** — not a new technique.

Reproducing the run

A2

FROZEN REPLAY — NO CREDENTIALS

Source the offline bundle's env and every LLM call is served from a curated cache. A miss hard-fails rather than silently spending money.

```
source .repro/scripts/activate_repro_env.sh
```

LIVE RE-RUN

```
python run_experiment.py --backend modal --tid state_ranker_v10_lgbm_blindset_B --eval_dataset blindset_B
```

ARTIFACTS

Models and data: [hf.co/datasets/Npatta01/music-crs-repro-2026](https://huggingface.co/datasets/Npatta01/music-crs-repro-2026)